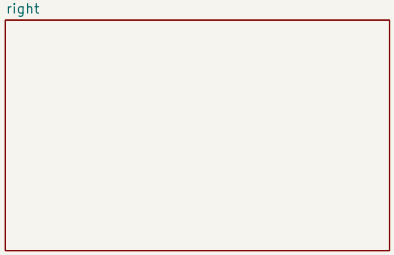
Keygen: one PCB carrying both halves joined by snap-off tabs.
Left half = references 1xx, right half = 2xx. Each half has its own nets (separate GND, VCC).

left



File: left.kicad_sch

right



File: right.kicad_sch

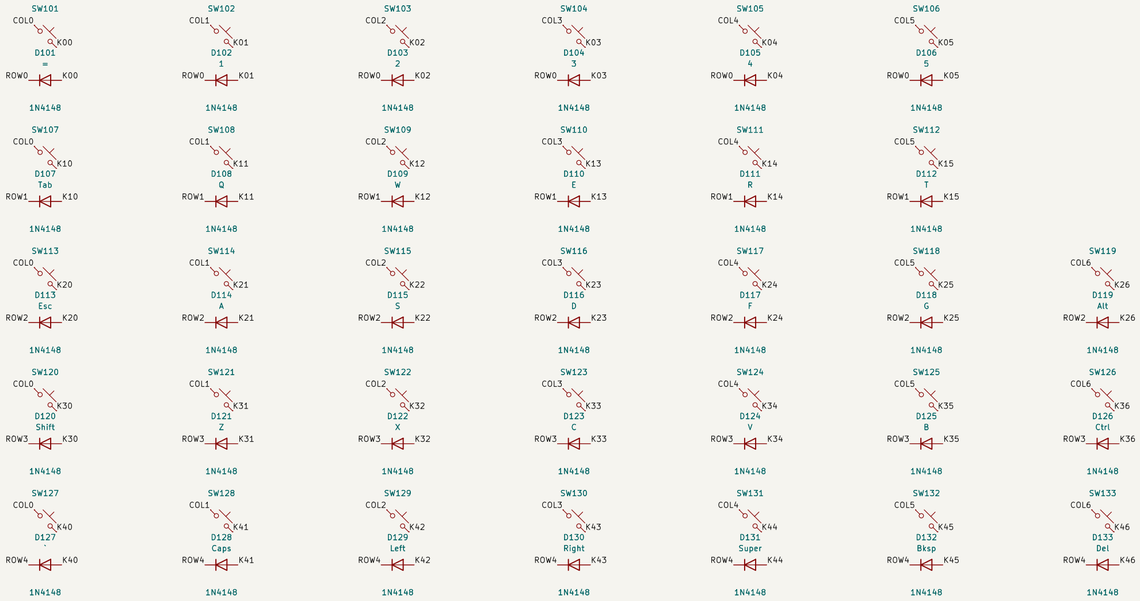
Generated by tools/gen_schematic.py — edit the script, not this file
github.com/milindmadhukar/keygen

Sheet: /
File: keygen.kicad_sch

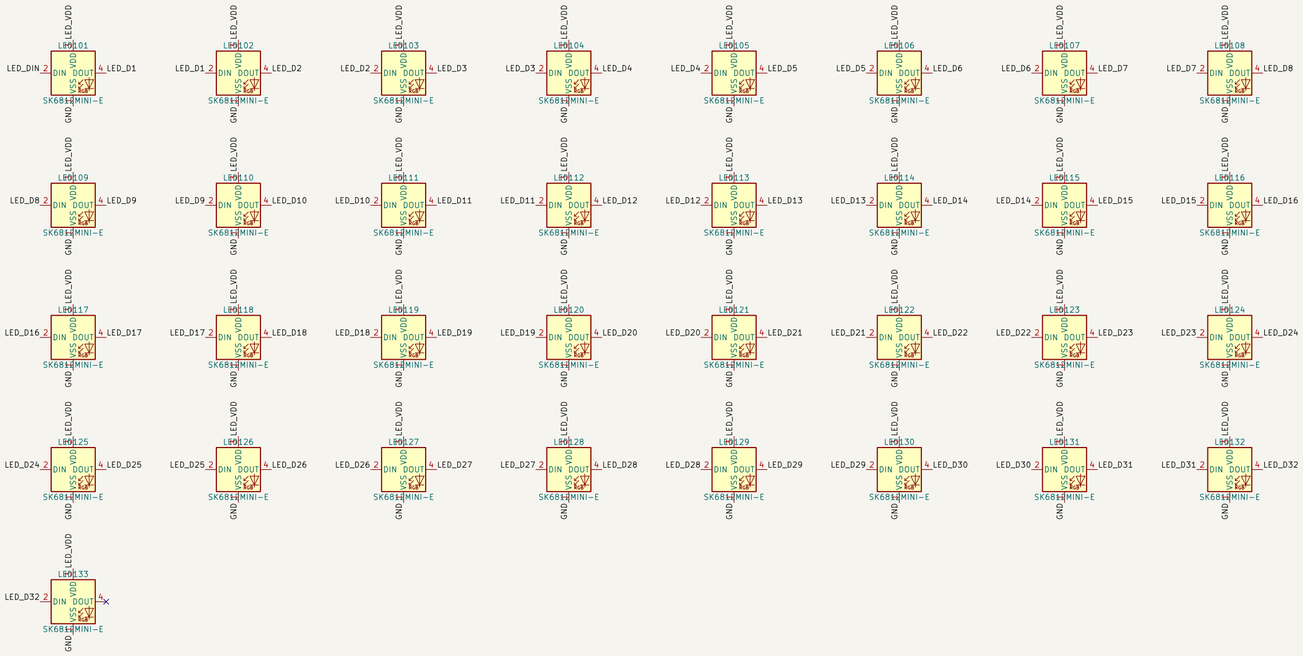
Title: Keygen

Size: A4	Date: 2026-10-20	**Rev: 0.1**
KiCad E.D.A. 10.0.7		Id: 1/3

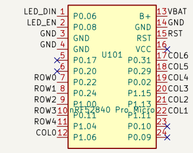
KEY MATRIX (COL2ROW: switch to COLn, diode cathode to ROWn)



PER-KEY LEDs (33) — powered from the battery through the LED power switch (Q2, LED_VDD)

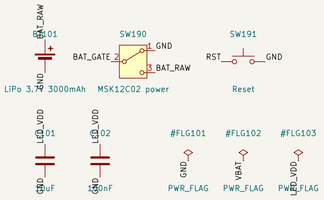


CONTROLLER (mounted on the PCB underside, USB-C to the top edge)



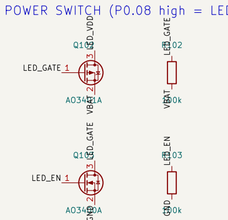
P0.08: LED power switch; VCC (pad 16) unused.
Spare: P0.17 (full speed), P0.20, P0.09, P0.10

POWER



C1/C2 sit at the LED power switch's output.

LED POWER SWITCH (P0.08 high = LEDs on)



BATTERY SWITCH (P-MOSFET; the slide switch only drives the gate)



Switch 2-1: gate to GND, on. Switch 2-3: gate to source, off.
R1 holds the gate off between positions. Leave the switch on while charging:
off, USB charges the cell through the body diode (~0.35 V at 500 mA).

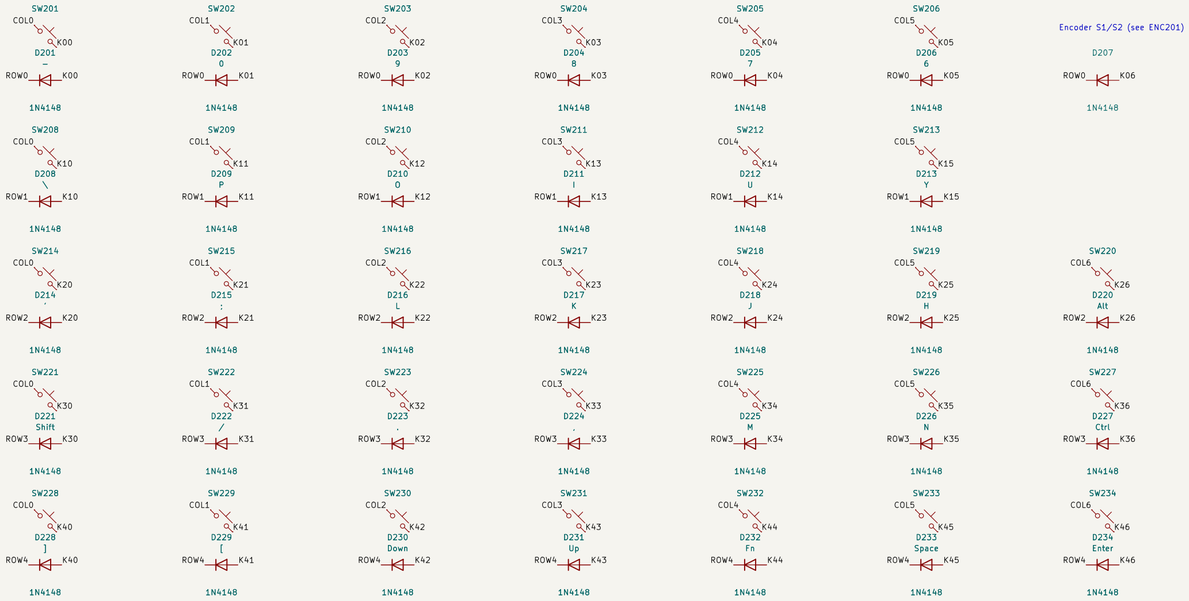
MOUNTING HOLES (M2)



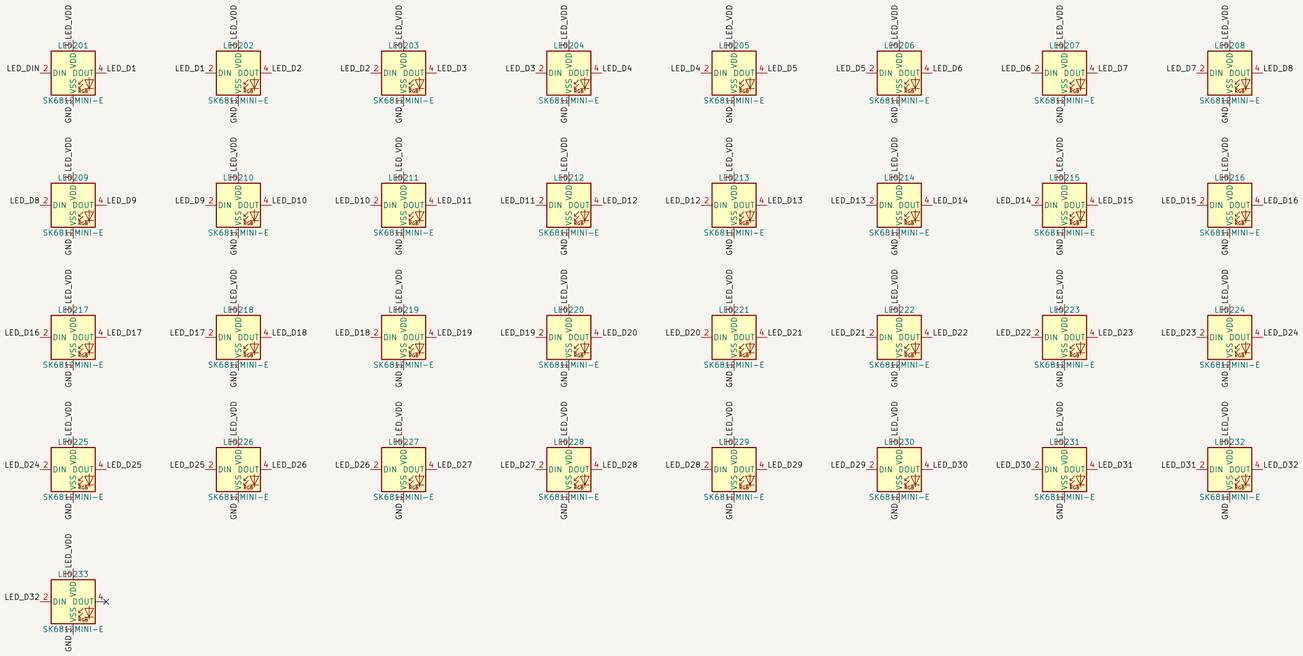
Generated by tools/gen_schematic.py — edit the script, not this file
github.com/milindmadhukar/keygen

Sheet: /left/		
File: left.kicad_sch		
Title: Keygen — left half		
Size: A2	Date: 2026-10-19	**Rev: 0.1**
KiCad E.D.A. 10.0.7		Id: 2/3

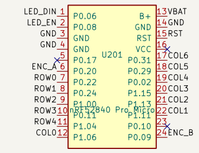
KEY MATRIX (COL2ROW: switch to COLn, diode cathode to ROWn)



PER-KEY LEDs (33) – powered from the battery through the LED power switch (Q2, LED_VDD)

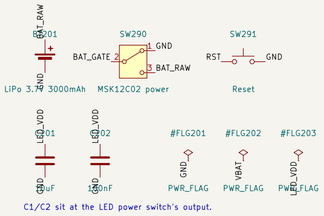


CONTROLLER (mounted on the PCB underside, USB-C to the top edge)



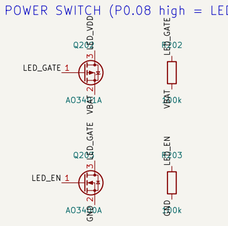
Encoder B on P0.09 (NFC pin): set nfct-pins-as-gpios.
P0.08: LED power switch. VCC (pad 16) unused.
Spare: P0.17 (full speed), P0.10

POWER



C1/C2 sit at the LED power switch's output.

LED POWER SWITCH (P0.08 high = LEDs on)

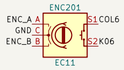


BATTERY SWITCH (P-MOSFET; the slide switch only drives the gate)



Switch 2–1: gate to GND, on. Switch 2–3: gate to source, off.
R1 holds the gate off between positions. Leave the switch on while charging:
off, USB charges the cell through the body diode (~0.35 V at 500 mA).

ROTARY ENCODER (EC11) – push switch in the matrix at ROW0/COL6



MOUNTING HOLES (M2)



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github.com/milindmadhukar/keygen

Sheet: /right/
File: right.kicad_sch

Title: Keygen – right half

Size: A2	Date: 2026-10-19	**Rev: 0.1**
KiCad E.D.A. 10.0.7		Id: 3/3